INCH-POUND
MIL-R-23837B(SH)
04 May 1992
SUPERSEDING
MIL-R-23837A(SHIPS)
20 December 1965
(See 6.9)

MILITARY SPECIFICATION

RADIAC SET AN/PDR-65( )

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1.1 Scope. This specification covers radiac set AN/PDR-65( ) which is a portable, watertight device measuring gamma radiation rates between 1 and 10,000 rad per hour (rad/h) and integrating doses up to 10,000 rad.

2 APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6665
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:04Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

MILITARY

MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.

MIL-M-10304 - Meters, Electrical Indicating, Panel Type, Ruggedized.

MIL-M-10304/18 - Meters, Multiscale, D.C, Panel Type, Ruggedized (Round Flange, 3-1/2 inch, Style 36).

MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Provisioned Items (Repair Parts): Packaging of.

STANDARDS

MILITARY

MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.

MIL-STD-109 - Quality Assurance Terms and Definitions.

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

MIL-STD-2036 - General Requirements for Electronic Equipment Specifications.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

PUBLICATIONS

ASSISTANT SECRETARY OF THE NAVY (S&L)

NAVMAT P4855-1 - Navy Power Supply Reliability and Design and Manufacturing Guidelines.

NAVAL AIR SYSTEMS COMMAND (NAVAIR)

WS-6536 - Process Specification Procedures and Requirements for Preparation and Soldering of Electrical Connections.

(Application for copies should be addressed to the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:04Z
Check the source to verify that this is the current version before use.

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
D 3951 - Standard Practice for Commercial Packaging. (DOD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.5) in accordance with 4.4.

3.2 General. The equipment shall be in accordance with MIL-STD-2036 except as otherwise specified herein.

3.3 Reliability. The equipment shall have a mean-time-between failure (MTBF) of 5000 hours (see 6 3 and appendix A).

3.4 Maintainability. The equipment shall have a mean-time-to-repair (MTTR) of not greater than 30 minutes (see 6.3 and appendix B).

3.5 Design requirements. Design requirements shall be as follows.

3 5.1 Composition. The instrument shall consist of a detector and associated circuitry, dose rate measuring circuits, digital dose register, audible sound, dosemeasuring circuit, carrying harness, bulkhead mounting accessory, twisted pair or multiconductor line (200 feet), and carrying case.

3 5.1 1 Construction (modular). The requirements for modular construction are not applicable.

3.5.1.2 Soldering and soldering processes. Soldering, soldering procedures, and processes shall conform to NAVAIR WS-6536.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:04Z
Check the source to verify that this is the current version before use.

3.5.2 Environmental conditions. The equipment shall meet the accuracy requirements of 3.6 when operated under the following conditions as specified in MIL-STD-2036 or as modified herein:

(a) Temperature, range 3.
(b) Humidity.

3.5.3 Electrical. The instrument, powered by four C-size nickel-cadmium rechargeable cells, shall include a separate watertight compartment accessible for battery replacement. A power supply, provided to recharge the cells, shall operate from 110 volts alternating current.

3.5.3.1 Connectors. External connectors shall be used in accordance with MIL-STD-2036.

3.5.3.2 Fuse indicators. Blown fuse indicators are not required.

3 5.4 Maintenance Test points and test features are required. The test points shall be accessible.

3 5.5 Mechanical. Mechanical requirements shall be as follows.

3.5.5.1 Shock. Shock requirements shall be for grade A, class I, lightweight, type A in accordance with MIL-S-901.

3.5.5.2 Enclosure. The degree of enclosure shall be watertight except that the power supply and remote control unit shall be dripproof in accordance with MIL-STD-108.

3 5.5.3 Vibration The equipment shall be subjected to the type I vibration test of MIL-STD-167-1 and shall be energized during the test. The instrument shall operate satisfactorily without internal adjustments and without replacements of any parts. There shall be no evidence of collision between any parts of the equipment.

3.5.6 Controls, indicators, and panel layout. Controls, indicators, and panel layout requirements shall be as follows:

(a) Tactile identification of controls is not required.
(b) Locking devices on controls are not required.
(c) Elapsed time meters are not required.

3.5.7 Technical information Instruction plates are not required (see 6.4).

3.5.8 Description. The radiac set shall consist of a watertight, portable, battery-operated radiacmeter; carrying case; shoulder strap; and bulkhead mounting accessory.

3.6 Performance and detail requirements. Performance and detail requirements shall be as follows:

3.6.1 Range. Range requirements shall be as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:04Z
Check the source to verify that this is the current version before use.

3.6.1.1 Doserate function. The instrument shall measure gamma doserate over the following ranges: 0-10, 100, 1,000, 10,000 rad/h.

3.6.1.2 Dose measuring function. The instrument shall integrate the gamma dose in increments of 1 rad up to a total of 9999 rad, at rates of 10,000 rad/h or less. Detector leakage shall contribute less than one extra count (1 rad) for each 250 hours of operation.

3.6.1.3 Accuracy. In determining accuracy, gamma energy dependence and directional dependence of the instrument shall be considered and treated as a combined effect. Accuracy shall be determined at 5 or more different photon energies between 70 kiloelectronvolts (keV) and 1.3 megaelectronvolts (MeV). Ca^{137} and Co^{60} shall be two of the isotopes used to obtain these energies.

3.6.1.4 Doserate accuracy. In the calibrating direction the indicated doserate shall be within plus or minus 20 percent of the true values for the range of energies specified above, with the exception that the doserate error shall be less than plus or minus 40 percent on the 0-10,000 rad/h range.

3.6.1.5 Dose accuracy. In the calibrating direction the integrated dose shall be within plus or minus 20 percent of the true dose for the range of energies specified above, up to 1000 rad/h, and plus or minus 40 percent between 1000 and 10,000 rad/h. For doserate in excess of 10,000 rad/h the dosimeter section shall continue to integrate at a rate of not less than 10,000 rad/h plus or minus 40 percent.

3.6.1.6 Directional dependence. The instrument shall be tested for directional dependence about 3 mutually perpendicular axes of the equipment.

3.6.1.6.1 Indicated doserate and dose. For any combination of energy and direction tested according to 3.6.1, the indicated doserate and dose shall be within minus 60 percent to plus 40 percent.

3.6.1.6.2 Average indicated doserate and dose. For any energy tested, the average indicated dose and doserate about any of the three axes of rotation shall be within plus or minus 30 percent of the true values.

3.6.1.6.3 Indicated doserate and dose (30 keV). For a source energy of 30 keV, the indicated dose and doserate shall be less than one-half the true values, with negligible response to energies less than 15 keV.

3.6.1.7 Saturation effects

3.6.1.7.1 Rate meter indication. When the instrument is subjected to fields in excess of 10,000 rad/h, the rate meter shall indicate not less than the indication for a true field of 10,000 rad/h, subject to the accuracy limitations in 3.6.1.

3.6.1.7.2 Dose measuring function. When subjected to fields greater than 10,000 rad/h, the dose measuring function shall continue to integrate at a rate not less than that for a true field of 10,000 rad/h, subject to the accuracy limitations in 3.6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:04Z
Check the source to verify that this is the current version before use.

3.6.1.8 Size. The radiacmeter shall be of a rectangular form capable of being hand carried or suspended by shoulder straps at the side. The volume shall be less than 200 cubic inches and no dimension shall exceed 9 inches.

3.6.1.9 Weight. The radiacmeter, including batteries, shall weigh no more than 7 pounds.

3.6.1.10 Electromagnetic compatibility. The radiac set shall meet the requirements of MIL-STD-461, part 5 (Class A4), tests RE02, RS01, and RS03. The limits for test RS03, however, shall be as in table I.

TABLE I. Frequency and field intensity for electromagnetic compatibility.

Frequency (MHz) 1/	Field intensity (V/m) 2/
0.014 - 0.30	1.0 average
3/ 0 30 - 1.5	1.0 average
3/ 1.5 - 30	10 average
30 - 12,400	1.0 average

1/ Megahertz (MHz).
2/ Volts per meter (V/m).
3/ Test at only three discrete frequencies within these ranges.

3 6.2 Detailed requirements

3.6.2.1 Radiacmeter. This instrument shall consist of a detector and associated circuitry, rate and dose measuring circuits, together with range changing switch, batteries, indicating meter, and digital dose register.

3.6.2.1.1 Circuitry. Transistor and other semiconductor components shall conform to MIL-STD-2036 Use of electron tubes shall be subject to the approval of the bureau or agency concerned.

3.6.2.1.2 Batteries. The instrument shall be powered by 4 C-size nickel-cadmium rechargeable cells and shall be capable of 40 hours of continuous operation in a field of 500 rad/h with the internal sounder on the 30 hours with 10 percent intermittent operation of dial lights. In fields greater than 500 rad/h battery life shall be long enough for 20,000 rad dose indication without dial light operation and 15,000 rad dose indication with 10 percent intermittent operation of dial light.

3 6 2.1.3 Battery compartment. The battery compartment shall be accessible from the outside for charging or changing the batteries without violating the watertight integrity of the compartment containing the electronic circuitry, indicators, and switch. A built-in connector shall be provided to accept a 2-wire cable from the battery charger.

3.6.2.1.4 Detector. The detector shall be of the recycling ion chamber type, argon filled and enclosing the electrometer tube, or of the G-M detector type. If an ion chamber is used as the detector, a guard ring shall separate the oppositely charged electrodes. The sensitivity of the detector shall be so that, with the circuitry of which it is a part, the basic detector shall recycle for each increment of dose equal to 0.5 millirad with provisions for varying the

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:04Z
Check the source to verify that this is the current version before use.

sensitivity by approximately plus or minus 40 percent. The output from the ion chamber shall be in the form of pulses suitable for driving a count integrator such as the "Incremag" manufactured by General Time Corporation or equal. The ratios of counts (Input/Output) shall be 2000:1, so that one count at the output of the counter shall be equal to the least count on the dose register; that is 1 rad. If the G-M Detector is used as the detector, its sensitivity shall be a minimum of 0.7 Counts/Min/Mr/Hr. Provisions shall be made to permit varying the calibration of the resultant readout by a minimum of plus or minus 20 percent.

3.6.2.1.5 Dose indicating register. The dose indicating register shall be a digital counter capable of performing the functions specified in 3.6.1.

3.6.2.1.6 Count rate circuit. Each time the detector recycles, a count rate metering circuit shall be triggered such that an indication on a current meter will be proportional to the gamma ray dose rate.

3.6.2.1.6.1 Count rate meter. The doserate meter shall be of the ruggedized, scale-changing type conforming to MIL-M-10304 and MIL-M-10304/18, and shall have a minimum full scale current of 20 microamperes.

3.6.2.1.7 Circuit time constant. The response time shall be defined as the length of time required for the meter pointer to reach 90 percent of the final indication after the sudden application of a constant radiation field The effective meter circuit time constant shall be such that the response time is 5 seconds plus or minus one second.

3 6.2.1.8 Remote capability The detector assembly shall be capable of being separated from the instrument housing and operated remotely. The requirement when operating remotely shall be the same as when the detector is integral with the instrument housing. The detector shall include the necessary circuitry to drive the signal through a minimum of 500 feet of twisted pair or multiconductor line when the detector is used in a remote location. In this case, power and control of the remote detector shall be provided by the basic unit.

3.6.2.1.9 Sounder radiacmeter. Provisions shall be made for an audible sound. For dose rates of 1 0 rad/h and below, each detector circuitry recycling event shall produce an audible "beep". For dose rates above 1.0 rad/h the "beep" rate shall not decrease.

3.6.2 1.10 Controls, radiacmeter. All controls and switches required for operation of the equipment shall be located on the top panel of the radiacmeter unless otherwise specified.

3.6.2.1.11 Control switch, radiacmeter. A single switch which shall control the radiacmeter operation shall have the following marked positions:

(a) Off-battery condition
(b) 10,000 rad/h.
(c) 1,000 rad/h.
(d) 100 rad/h.
(e) 10 rad/h.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:04Z
Check the source to verify that this is the current version before use.

3.6.2.1.12 Dosimeter rezeroing control. The radiacmeter shall have a control to rezero the digital register. The control shall be located so that it is accessible without disassembly of the equipment but it shall not protrude from the surface of the equipment. The mechanism to rezero the register shall be constructed so that accidental rezeroing is not feasible.

3.6.2.1.13 Illumination switch. An externally operated switch shall be provided for illumination of the visual indicators.

3.6.2.1.14 Calibration controls. Access to the meter calibration controls in the circuitry shall be provided by an entrance that does not require removal of the control panel. However, sensitivity adjustment to accommodate variations in G-M tube sensitivity may be internal. Calibration controls shall be provided to:

(a) Adjust the recycling detector sensitivity to a value which shall cause the register to indicate 1 rad on each output of the count integrator when a recycling detector is used.
(b) Adjust the sensitivity of the dose rate meter by means of a single control so that one adjustment shall calibrate all ranges
(c) When a G-M detector is used, calibration adjustment shall be provided to permit calibration of the dose indication. In addition, a control shall be provided to enable calibration of the analog meter indication of dose rate. Each of the above controls shall provide at least plus or minus 20 percent of indicated dose or dose rate.

3.7 Bulkhead mounting accessory. An accessory mounting bracket shall be provided for mounting the radiacmeter to a wall panel or bulkhead so that the instrument may be operated in a fixed manner from the internal battery supply or from an alternating current (ac) power supply. The bulkhead mounting accessory shall consist of the following

(a) Accessory mounting bracket.
(b) Power supply and remote control unit.
(c) Detector mounting bracket.

3.7.1 Bracket dimensions. Accessory mounting bracket shall not exceed 10 inches high by 6-1/2 inches deep by 10-1/2 inches wide.

3.7 2 Mounted dimensions. Power supply and remote control unit shall be contained in the same enclosure and mounted on the accessory mounting bracket. The size shall not exceed 4-3/4 inches wide by 4-1/2 inches deep by 7-1/2 inches high and the weight shall not exceed 4-1/4 pounds.

3.7.2.1 Power supply. The power supply shall conform to the requirements of NAVMAT P4855-1 and operate from a 115-volt, 60-cycle, single-phase line and shall provide the necessary direct current power to:

(a) Operate the radiacmeter with internal batteries installed in the radiacmeter so that operation is uninterrupted in event of ac power failure.
(b) Recharge the nickel-cadmium batteries in the radiacmeter in 24 hours or less and maintain a trickle charge after the batteries are charged.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:04Z
Check the source to verify that this is the current version before use.

(c) Operate the auxiliary sounder circuit.

3.7.2.1.1 Battery provisions. Provisions shall be made to insure that the batteries shall not be damaged when charged for an indefinite length of time.

3.7.2.1.2 Pilot light and fuse. A pilot light and fuse shall be provided. Blown fuse indicator shall not be required.

3.7.2.1.3 Power supply switch. A single switch shall be provided and it shall be marked as follows:

Power ON OFF.

3.7.2.2 Remote control unit. The remote control unit shall facilitate connecting two radiacmeters located up to 500 feet apart, so that the doserate at one radiacmeter can be monitored with the readout of the other radiacmeter, without affecting the monitoring of local dose at either radiacmeter.

3 7 2.2 1 Remote control unit switch A single switch shall control the doserate input signal to the instrument and it shall be marked as follows:

Detector:
(a) Local.
(b) Remote.

3.7.2.2.2 Auxiliary sounder. The remote control unit shall contain an auxiliary sounder which accepts input signals from either the local detector or remote detector. The remote control unit switch shall also control the doserate input signal to the auxiliary sounder.

3.7.2.2.3 Auxiliary sounder "beep". The auxiliary sounder shall produce an audible "beep" tone for each detector circuitry recycling event for doserates of 10 rad/h and below. For doserates above 10 rad/h, the "beep" repetition rate shall not decrease. The dominant frequency of the beep tone shall be between 1.0 kilohertz. At a distance of two feet, the sound pressure level shall be between 70 decibels (db) and 90 db using 0 db reference of 2 by 10^{4} dynes per square centimeter.

3.7.2.2.4 Sounder switch. A single switch shall be provided to turn the auxiliary sounder off and on, and it shall be marked as follows:

Sounder:
ON
OFF

3 7.3 Detector mounting bracket A separate bracket shall be provided for mounting the detector to a wall, panel or bulkhead. It shall be made of aluminum and its size shall be 2-9/16 inches by 6-3/4 inches by 2-13/16 inches.

3.8 Carrying-harness. The shoulder harness shall allow the radiacmeter to be slung from the shoulder and shall be capable of being quickly and easily attached to, or detached from the radiacmeter. The harness shall be fabricated of a nonporous material which resists absorption of radiactive water or other contaminated material.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:04Z
Check the source to verify that this is the current version before use.

3.9 Carrying case. Each radiac set AN/PDR-65( ) shall be provided with a splashproof, lightweight (aluminum) case designed to house the following:

(a) Radiacmeter.
(b) Carrying harness.
(c) Instruction books (two).
(d) Bulkhead mounting accessory.
(e) Twisted pair of multiconductor line (200 feet).
(f) Batteries.

3.9.1 Carrying case handle. Means shall be provided for securing the radiacmeter and accessories in the carrying case to prevent shifting the damage during shipping and storage. A handle or handles shall be provided for convenience in carrying and shall not interfere with orderly stacking and stowing of equipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements

4 1 1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program (see 6.3). The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Government verification. All quality and safety assurance operations performed by the contractor will be subject to Government verification at any time Verification will consist of (a) surveillance of the operations to determine that practice, methods, and procedures of the quality and safety program requirements are being properly applied, (b) Government inspection to measure quality and safety of product to be offered for acceptance, and (c) Government inspection of delivered items to ensure compliance with this specification not excluding any requirement of the specification for which detailed tests are not specified herein.

4.1.2.1 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:04Z
Check the source to verify that this is the current version before use.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.4).
(b) Quality conformance inspection (see 4.5).

4.3 Detector assembly (recycling detector). When a recycling detector is used, each detector shall be temperature cycled and tested for gas leaks before being installed in an equipment. The test procedure and temperature cycle shall be as follows:

(a) In a circuit with the same characteristics as the production radiacmeter, the detector units response to Cs^{137} or Co^{60} shall be determined.
(b) Decrease temperature to minus 62 ± 2 degrees Celsius (°C) and hold for at least 72 hours.
(c) Increase temperature to plus 75 ± 2°C and hold for at least 72 hours
(d) Check the detector assembly response in accordance with 4.3(a). The response shall not vary more than 5 percent from the reading taken as specified in 4.3(a).
(e) All detector units shall be stored for a period of 30 days and again checked for response in accordance with 4.3(a). The response shall not vary more than 5 percent from the reading taken as specified in 4.3(a).

4 3 1 Detector assembly (G-M detector) When a G-M detector is used, each G-M tube shall be tested for satisfactory starting voltage and plateau length in accordance with the military specification of the particular G-M tube used.

4 4 First article inspection Unless otherwise specified (see 6.2), three units shall be required for first article inspection. First article inspection shall consist of the examination and testing necessary to determine compliance

with the requirements of this specification and MIL-STD-2036, using the examination and tests of MIL-STD-2036 listed in 4.4.1 of this specification (see 6.3 and 6.7).

4.4.1 First article examination and tests The following examination and tests of MIL-STD-2036 shall be performed.

(a) Examination.
(b) Performance
(c) Temperature - (low) (performed using sheltered equipment - procedure II, basic cold.) (temperature RANGE 3).
(d) Temperature - (high) (perform with plus 50°C high temperature range) (temperature RANGE 3).
(e) Humidity.
(f) Salt fog.
(g) Shock (may be performed on one unit only).
(h) Vibration.
(i) Enclosure (to follow shock and vibration tests).
(j) Steady state voltage and frequency.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:04Z
Check the source to verify that this is the current version before use.

(k) Transient voltage.
(l) Transient frequency.
(m) Spike voltage.
(n) Power interruption.
(o) Electromagnetic environmental effects (includes test RE01, RE02, RS01, RS02 and RS03 of MIL-STD-462) (see table I for RS03).
(p) Inclination.
In addition, the following tests shall be performed to determine conformance with 3.6.1:
(q) Doserate function (3.6.1.1).
(r) Doserate measuring function (3.6.1.2).
(s) Accuracy (3.6.1.3).
(t) Doserate accuracy (3.6.1.4).
(u) Dose accuracy (3.6.1.5)
(v) Directional dependence (3.6.1.6).
(w) Saturation effects (3.6.1.7).

4.4.2 Non-standard parts. First article approval will not be made until all non-standard parts have been approved (see 6.7). Germanium semi-conductors are not acceptable as non-standard parts for this equipment.

4 5 Quality conformance inspection.

4.5.1 "Break-in" period. Prior to the production inspection, each equipment shall be operated under average operation conditions for a minimum of 90 hours. The ac power supply may be used for this "break-in" period

4.5.2 Quality conformance examinations and tests. Quality conformance inspections shall be made on every equipment and in addition to the inspections specified in MIL-STD-2036 include the following inspection:

(a) Accuracy (calibration direction using Cs^{137} or Co^{60} only) (see 3.6.1).

4.5.3 Quality conformance sampling. Quality conformance sampling shall conform to MIL-STD-2036 and table II. Quality conformance sampling shall be performed on equipment that has passed quality conformance inspection, and shall consist of the following examination and tests:

(a) Low temperature (performed using sheltered equipment - procedure II, basic cold) (temperature RANGE 3).
(b) High temperature (perform with plus 50°C high temperature range (temperature RANGE 3).
(c) Enclosure (to follow shock test).
(d) Examination.
(e) Humidity.
(f) Shock.
(g) Energy dependence.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:04Z
Check the source to verify that this is the current version before use.

TABLE II. Quality conformance sampling.

Lot size	Sample size	Accept	Reject 1/2/3/
2 - 15	2	0	1
16 - 25	3	0	1
26 - 90	5	0	1
91 - 150	6	0	1
151 - 280	7	0	1
281 - 500	9	0	1
501 - 1200	11	0	1
1201 - 3200	13	0	1
3201 and over	15	0	1

1/ All defective items must be replaced with acceptable items prior to lot acceptance.
2/ Inspection sample size until reject criteria is reached.
3/ Rejected lots may be screened and resubmitted for inspection and retest.

4.6 Disposition of shock tested equipment The shock tested sample shall be thoroughly reconditioned by the contractor before resubmitting it as production equipment.

4.7 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing and marking for shipment, stowage, and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.6.)

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts.

5.1.1 Basic equipment or item.

5.1.1.1 Preservation Preservation shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation and may conform to the contractors commercial practice when such meets these requirements.

5.1.1.2 Packing. Packing shall be accomplished in a manner which will ensure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to ASTM D 3951 as applicable to the mode of transportation and may conform to the contractors commercial practice when such meets these requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:04Z
Check the source to verify that this is the current version before use.

5.1.1.3 Marking. Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, national stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.1.2 Onboard repair parts. Onboard repair parts shall be preserved, packed, and marked as specified in MIL-E-17555.

5.2 Domestic shipment and storage or overseas shipment. The requirements and levels of preservation, packing, and marking for shipment shall be as specified by the contracting activity (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6 1 Intended use. This specification is intended for the procurement of portable radiac sets, AN/PDR-65( ), used for measuring gamma radiation rates between 1 and 10,000 rad/h and integrating doses up to 10,000 rad.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2 2).
(c) When first article inspection is required (see 3.1)
(d) When first article inspection quantities are other than as specified (see 4.4).
(e) Levels of preservation, packing, and marking (see 5.2).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.3 and appendix A	DI-R-7079	Reliability program plan	----
3.3 and appendix A	DI-R-7082	Reliability predictions report	----
3.3 and appendix A	DI-RELI-80252	Reliability test reports	----
3.4 and appendix B	DI-R-5189	Maintainability prediction data	----
3.4 and appendix B	DI-R-2129	Plan maintainability	----

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:04Z
Check the source to verify that this is the current version before use.

		demonstration	
4.1 1	DI-QCIC-81110	Inspection and test plan	----
4.4	DI-NDTI-80809	Test/inspection reports	Subtitle CDRL "Firt Article Inspection Report"

The above DID's were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract The technical manuals must be acquired under separate contract line item in the contract

6.4.1 In addition to the requirements of the technical manual specification, the contractor should determine the parts that can be replaced without effecting the calibration and list all such parts under the section titled "Maintenance".

6.5 First article When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the items(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.6 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped

6.7 NAVSEA approval and direction. Deviations from specified materials, procedures, and requirements and selection of specific alternative materials and procedures require NAVSEA approval or direction. Requests should include supporting documentation.

6.7.1 Non-standard parts. The first article inspection report accompanying the first article should include a list of non-standard parts approved or pending approval for the equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:04Z
Check the source to verify that this is the current version before use.

6.8 Subject term (key word) listing.

Dose rate
Dose register
Dosimeter
Gamma radiation
G-M detector
Recycling detector

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 6665-N592)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:04Z
Check the source to verify that this is the current version before use.

APPENDIX A

RELIABILITY

10. SCOPE

10.1 Scope. This appendix covers the information and procedures necessary to ensure that the reliability requirements are met. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS

20.1 Government documents.

20.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

MILITARY

MIL-STD-781 - Reliability Testing for Engineering Development, Qualification, and Production.
MIL-STD-785 - Reliability Program for Systems and Equipment Development and Production.

HANDBOOKS

MILITARY

MIL-HDBK-217 - Reliability Prediction of Electronic Equipment.
MIL-HDBK-781 - Reliability Test Methods, Plans, and Environments for Engineering Development, Qualification, and Production.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

30. RELIABILITY

30.1 Quantitative reliability requirements. The AN/PDR-65( ) Radiac Set shall have a specified mean-time-between-failure (MTBF) of 5000 hours.

30.2 Reliability program plans. The contractor shall prepare for approval a reliability program in accordance with the requirements of the procurement and the requirements of MIL-STD-785. The reliability program shall describe the tasks to be performed and the procedures for conducting and controlling the reliability program.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:04Z
Check the source to verify that this is the current version before use.

30.3 Reliability prediction. A design reliability prediction shall be performed in accordance with MIL-STD-785 (task 203) and MIL-HDBK-217. The prediction shall be based on the requirements of this specification. A functional block diagram and reliability block shall be submitted with the reliability prediction report. Mechanical parts shall be included as shown in paragraph 5.0, MIL-HDBK-217. The source and applicability of all failure data used for the prediction shall be documented and justified prior to use. The results of the prediction shall be compared to the specified MTBF and where the predicted values do not achieve the requirements, the contractor shall accomplish such changes as are necessary to increase the predicted values sufficiently to achieve the requirements.

30.4 Reliability testing. The reliability of the AN/PDR-65( ) Radiac Set shall be demonstrated on the initial production units in accordance with the requirements of MIL-STD-781 and MIL-HDBK-781. Failure shall be defined as any malfunction or parameter deviation that prevents the equipment from performing within the requirements set forth in this specification. Debugging and burn-in prior to the reliability test shall be limited to the debugging and burn-in to be performed on all production equipment prior to delivery to the Government. No preventative maintenance shall be performed during the reliability test. Reliability testing shall include vibration tests.

30.5 Performance parameters for reliability testing The parameters to be monitored shall include all of the requirements specified in section 3 of this specification.

30 6 Reliability qualification tests. Reliability qualification testing shall be conducted on the initial production units in accordance with MIL-STD-781 and MIL-HDBK-781. Testing shall be performed with test plan IV with the following test conditions:

Temperature	50 to 55°C (122 to 131 degrees Fahrenheit (°F)).
Temperature Cycling	None.
Vibration	2.2G ± 10 percent peak acceleration at any nonresonant frequency between 20 and 60 cycles per second (c/s) measured at the mounting points on the equipment. (Equipment designed for use on Navy ships shall be vibrated at 1G ± 10 percent. If the equipment is designed to meet a vibration requirement less severe than the requirement contained herein, the vibration may be reduced to the level specified as a design requirement.) The duration of vibration shall be at least 10 minutes during each hour of equipment operating time. When the equipment being tested contains circuit boards or cards, vibration shall be normal to the plane of the majority of the cards. This vibration requirement may be met by

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:04Z
Check the source to verify that this is the current version before use.

	mounting the equipment solidly (that is, without the equipment shock vibration mounts) to a strong flat plate which is supported by vibration isolators. The desired amplitude and frequency may be obtained by a simple vibration machine with an asymmetric weight on a shaft attached to the plane and driven by a motor. It is recommended that the vibration equipment be checked for proper operation each 24 hours of operation, and that the vibration transducer be "on" and monitored continuously during vibration.
Equipment On-Off Cycling	Turn on and let temperature stabilize, hold for 3 hours, then turn off and let temperature stabilize. This cycle shall continue throughout the test.
Input Voltage	Nominal specified voltage plus 5 percent to minus 2 percent.
Input Voltage Cycling	When so directed by the contracting activity, voltage cycling shall be accomplished as follows: The input voltage shall be maintained at 110 percent nominal for one-third of the equipment "on" cycle, and at 90 percent for the final one-third of the equipment "on" cycle This cycle procedure to be repeated continuously throughout the reliability test.
Input Voltage	Nominal specified voltage plus 5 percent to minus 2 percent.

Qualification testing shall be performed on three production units. Reliability testing shall be successfully completed prior to equipment shipment.

30 7 Reliability sampling tests. Reliability sampling tests shall be conducted throughout production to insure continued production of acceptable equipment. Testing shall be performed with test plan IV with the following conditions:

Temperature	50 to 55°C (122 to 131°F)
Temperature Cycling	None.
Vibration	2 2G ± 10 percent peak acceleration at any nonresonant frequency between 20 and 60 c/s measured at the mounting points on the equipment. (Equipment designed for use on Navy ships shall be vibrated at 1G ± 10 percent. If the equipment is designed to meet a vibration requirement less severe than the requirement contained herein, the vibration may be

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:04Z
Check the source to verify that this is the current version before use.

	reduced to the level specified as a design requirement.) The duration of vibration shall be at least 10 minutes during each hour of equipment operating time. When the equipment being tested contains circuit boards or cards, vibration shall be normal to the plane of the majority of the cards. This vibration requirement may be met by mounting the equipment solidly (that is, without the equipment shock vibration mounts) to a strong flat plate which is supported by vibration isolators. The desired amplitude and frequency may be obtained by a simple vibration machine with an asymmetric weight on a shaft attached to the plane and driven by a motor. It is recommended that the vibration equipment be checked for proper operation each 24 hours of operation, and that the vibration transducer be "on" and monitored continuously during vibration
Equipment On-Off Cycling	Turn on and let temperature stabilize, hold for 3 hours, then turn off and let temperature stabilize. This cycle shall continue throughout the test
Input Voltage	Nominal specified voltage plus 5 percent to minus 2 percent
Input Voltage Cycling	When so directed by the contracting activity, voltage cycling shall be accomplished as follows. The input voltage shall be maintained at 110 percent nominal for one-third of the equipment "on" cycle, and at 90 percent for the final one-third of the equipment "on" cycle. This cycle procedure to be repeated continuously throughout the reliability test.

MIL-STD-781 and MIL-HDBK-781 shall be used to determine the number of units to be selected for reliability sampling tests. Reliability testing for each lot shall be successfully completed prior to equipment shipment

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:04Z
Check the source to verify that this is the current version before use.

APPENDIX B

MAINTAINABILITY

10. SCOPE

10.1 Scope. This appendix covers the information and procedures necessary to ensure that the maintainability requirements of this specification are met. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance

20. APPLICABLE DOCUMENTS

20.1 Government documents.

20.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

MILITARY

MIL-STD-471 - Maintainability Verification/Demonstration/ Evaluation.

HANDBOOKS

MILITARY

MIL-HDBK-472 - Maintainability Prediction

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

30. MAINTAINABILITY

30.1 Quantitative maintainability requirements. The AN/PDR-65( ) Radiac Set shall have an equipment mean-time-to-repair (MTTR) of not greater than 30 minutes.

30.2 Maintainability prediction. Maintainability prediction data shall be prepared in accordance with MIL-HDBK-472 (see 6.3).

30.3 Maintainability demonstration plan. The detailed maintainability test plan shall be in accordance with MIL-STD-471.

30.4 Maintainability demonstration. The maintainability requirements for the equipment shall be successfully demonstrated in accordance with method III of MIL-STD-471. The maintainability test shall be performed on initial production units.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:04Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1 The preparing activity must complete blocks 1, 2, 3, and 8 In block 1, both the document number and revision letter should be given

2 The submitter of this form must complete blocks 4, 5, 6, and 7

3 The preparing activity must provide a reply within 30 days from receipt of the form

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements

I RECOMMEND A CHANGE:	1 DOCUMENT NUMBER	2 DOCUMENT DATE (YYMMDD)
	MIL-R-23837B(SH)	92/05/04

3. DOCUMENT TITLE RADIAC SET AN/PDR-65 ( )

4 NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible Attach extra sheets as needed)*

5 REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8 PREPARING ACTIVITY

a NAME Technical Point of Contact (TPOC): Mr. Ron Rhodes (SEA 04RL) PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	b TELEPHONE (Include Area Code) (1) Commercial TPOC: 703-602-2753 (2) AUTOVON 8-332-2753
c ADDRESS (Include Zip Code) Commander, Naval Sea Systems Command Department of the Navy (SEA 55Z3) Washington, DC 20362-5101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD Form 1426, OCT 89 *Previous editions are obsolete* 198-290

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:04Z
Check the source to verify that this is the current version before use.